

Question Four: Normalization**(10marks)****a) Describe the first three forms of Normalization.****(3marks)****b) Normalize the following table and list all the resulting tables.****(7marks)**

OrderID	CustomerID	CustomerName	ProductID	ProductName	Quantity	UnitPrice
0001	C001	James	P001	Laptop	2	500
0001	C001	James	P002	Mouse	3	20
0002	C002	Sarah	P003	Printer	1	200

Question Five: MySQL Queries**(10marks)**

A school has developed a Library Management Database to manage books borrowed by students. The database contains the following tables:

- STUDENT (StudentID (PK), FullName, Gender, Class)
- BOOK (BookID (PK), Title, Category, Author, CopiesAvailable)
- BORROW (BorrowID (PK), StudentID, BookID, BorrowDate, ReturnDate, FineAmount)

Assume the tables already contain data.

Answer the following SQL questions.

a) Create a new table named LIBRARIAN with the following fields:**(2marks)**

- LibrarianID (Primary Key)
- FullName
- Email
- BookID (Foreign Key referencing BOOK table)
- HireDate

b) Write an SQL statement to update the category of the book titled "Introduction to Databases" to "Computer Science".**(1mark)****c) Write an SQL query to display the total fine collected for each student. Display only students whose total fine is greater than 5,000 RWF.****(2marks)**